Mechanical & Aersopace Engineering: The Many Facets of Energy

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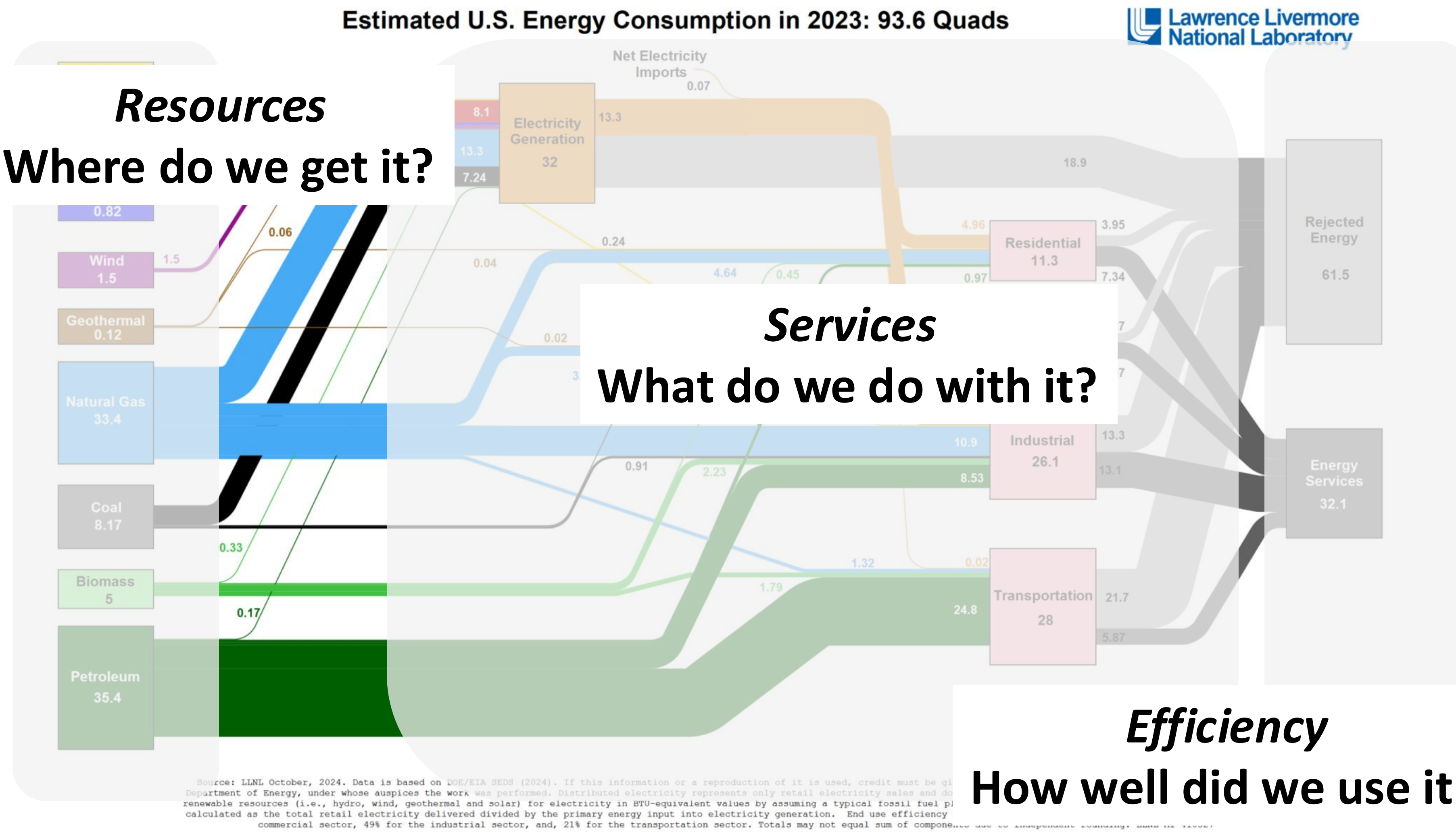
Energy Flows



Resources

Where do we get it?

Services

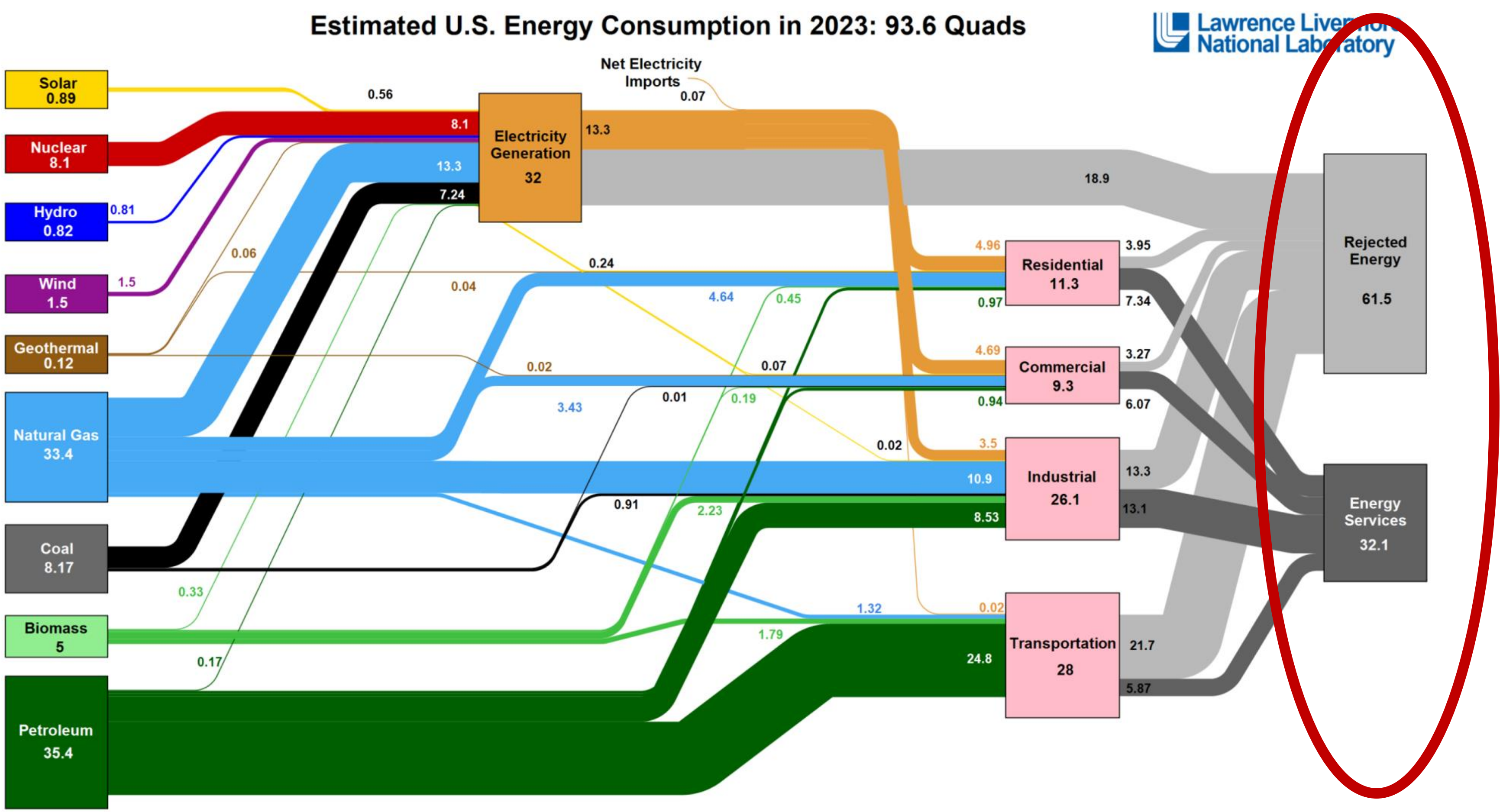
What do we do with it?

Efficiency

How well did we use it?

https://flowcharts.llnl.gov/commodities/energy

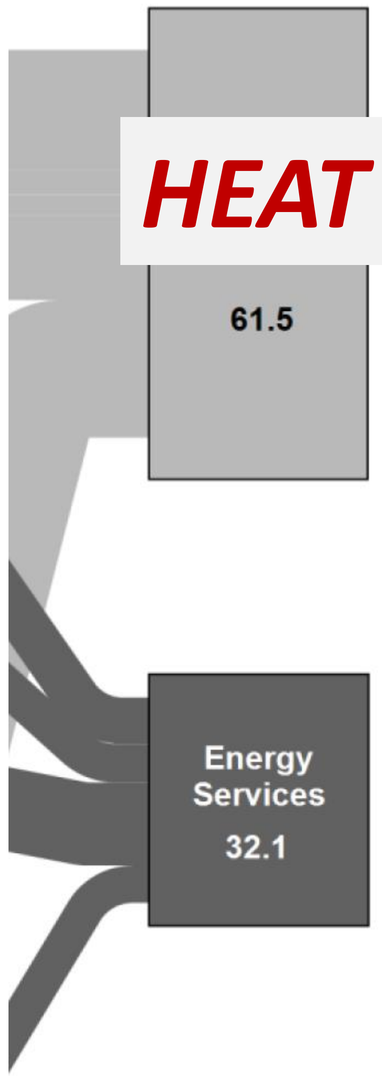
Where did all that energy go?



Source: LLNL October, 2024. Data is based on DOE/EIA SEDS (2024). If this information or a reproduction of it is used, credit must be given to the Lawrence Livermore National Laboratory and the Department of Energy, under whose auspices the work was performed. Distributed electricity represents only retail electricity sales and does not include self-generation. EIA reports consumption of renewable resources (i.e., hydro, wind, geothermal and solar) for electricity in BTU-equivalent values by assuming a typical fossil fuel plant heat rate. The efficiency of electricity production is calculated as the total retail electricity delivered divided by the primary energy input into electricity generation. End use efficiency is estimated as 65% for the residential sector, 65% for the commercial sector, 49% for the industrial sector, and, 21% for the transportation sector. Totals may not equal sum of components due to independent rounding. LLNL-MI-410527

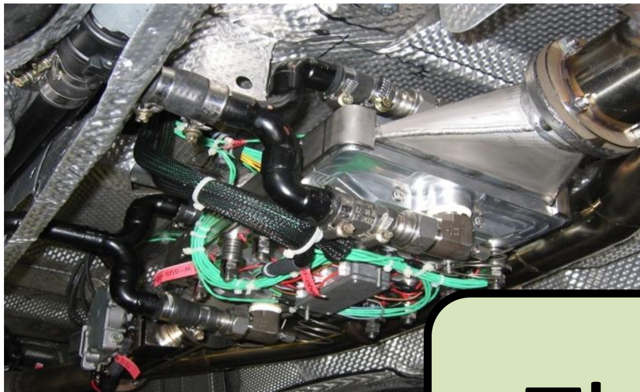
https://flowcharts.llnl.gov/commodities/energy

Where did all that energy go?



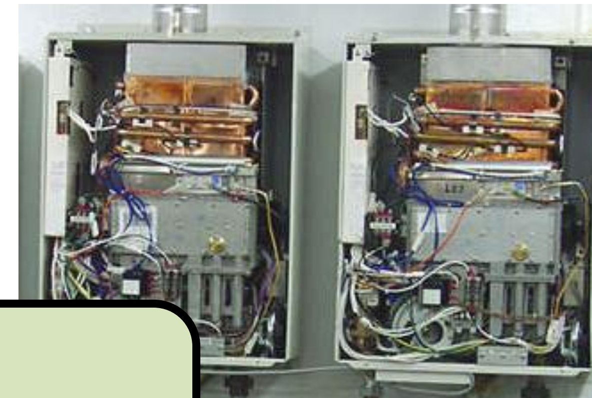
Energy Systems

Automotive Exhaust



www.bosch.com

Water Heater

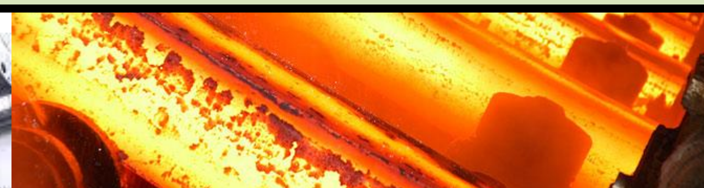


www.water.arredemo.org

Thermal Systems

Gas Turbine

www.usa.siemens.com



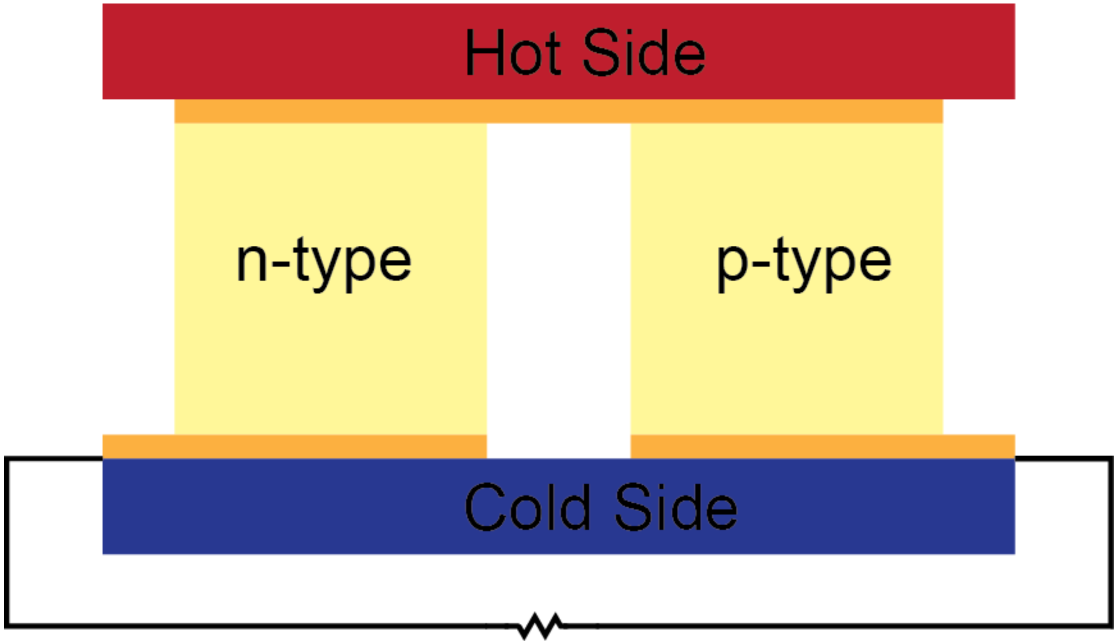
www.risenterprises.com

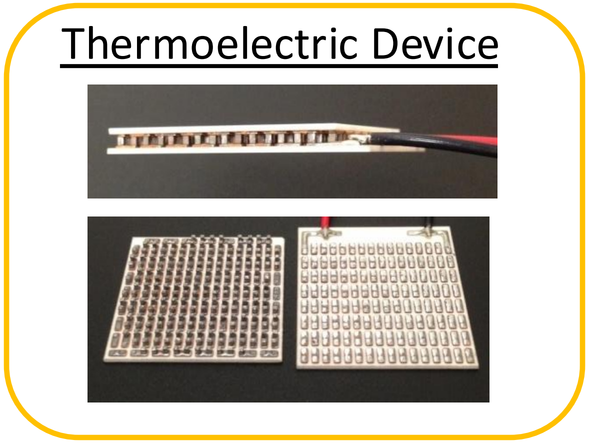
Glass Lehr

www.glassfurnacemaker.com

Thermoelectric Energy Conversion

Power generation: Convert thermal energy to electrical energy.





Advantages:

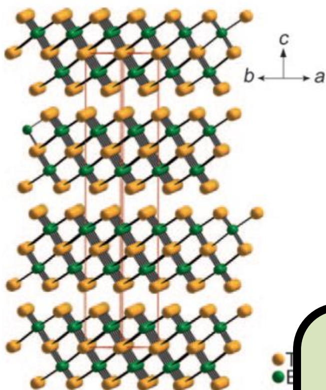
Distributed generation
Localized control of energy transfer
Reliable
No moving parts
Silent

Challenges:

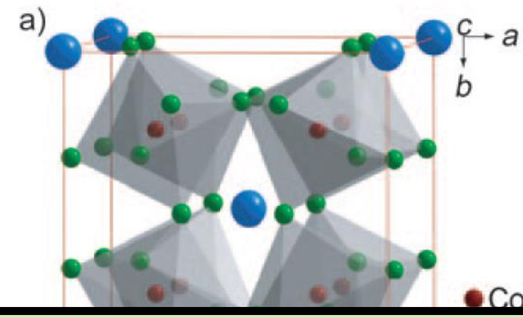
System integration
Conversion efficiency
Operating temperatures

Thermoelectric Materials

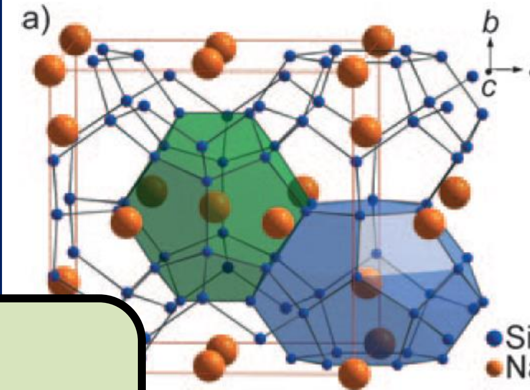
Chalcogenide



Skutterudite

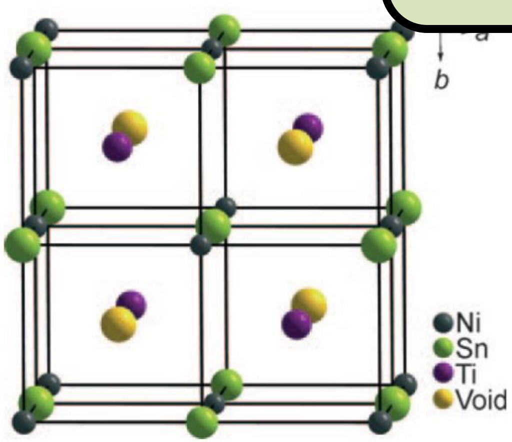


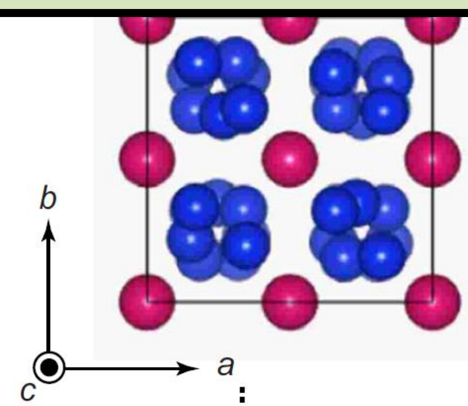
Clathrate



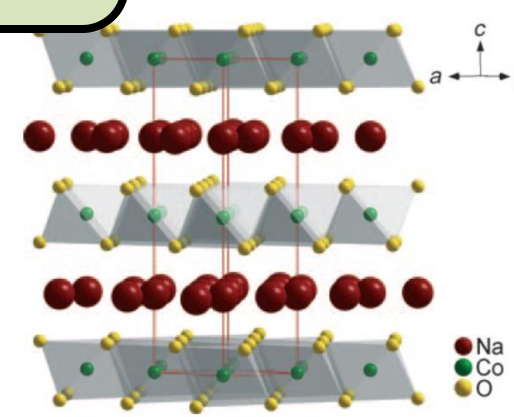
Materials Science

Half Heusler





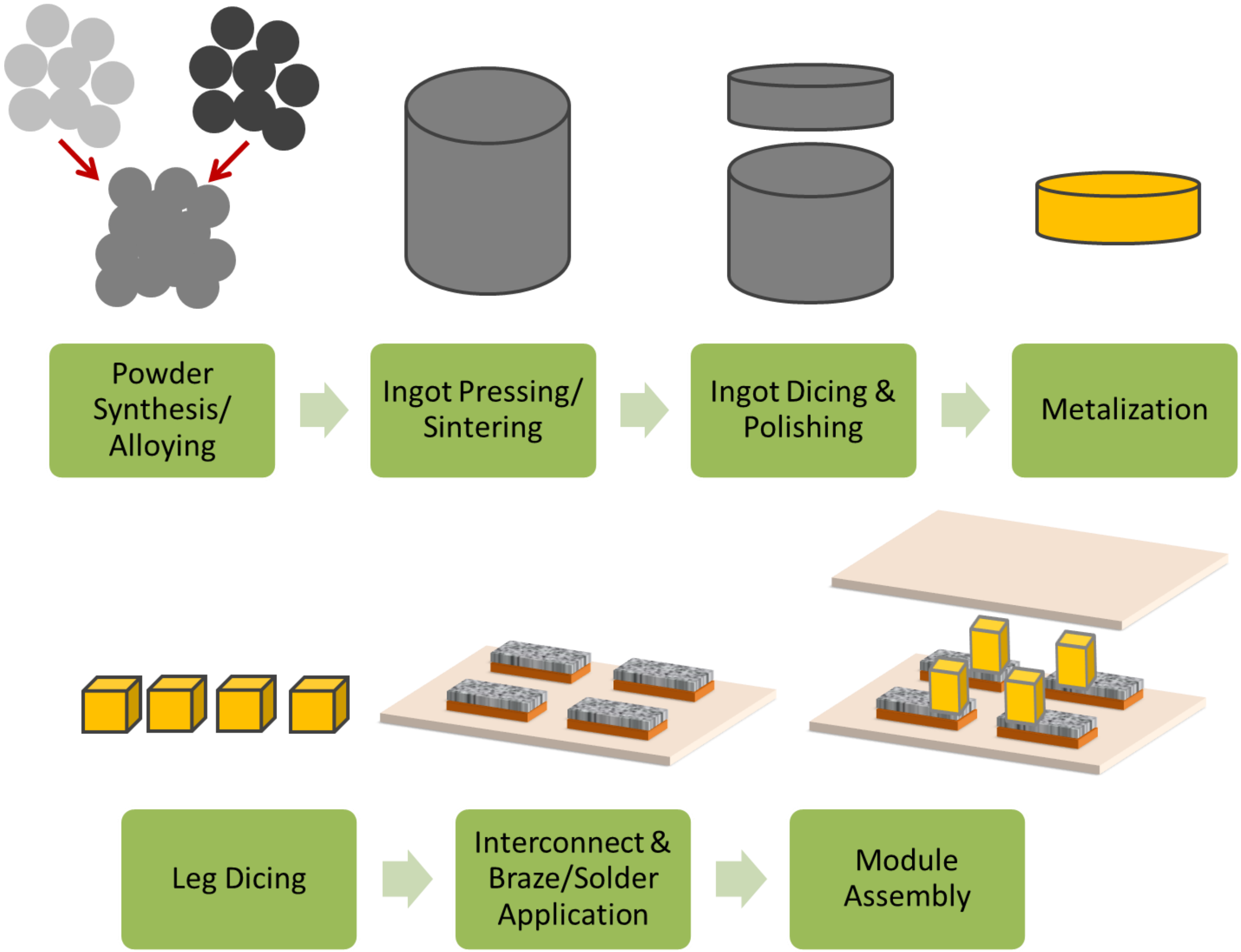
Metal Oxide



Sootsman *et al.*, *Angew .Chem.* (2009)

Miyazaki *et al.*, *Phys. Rev. B* (2008)

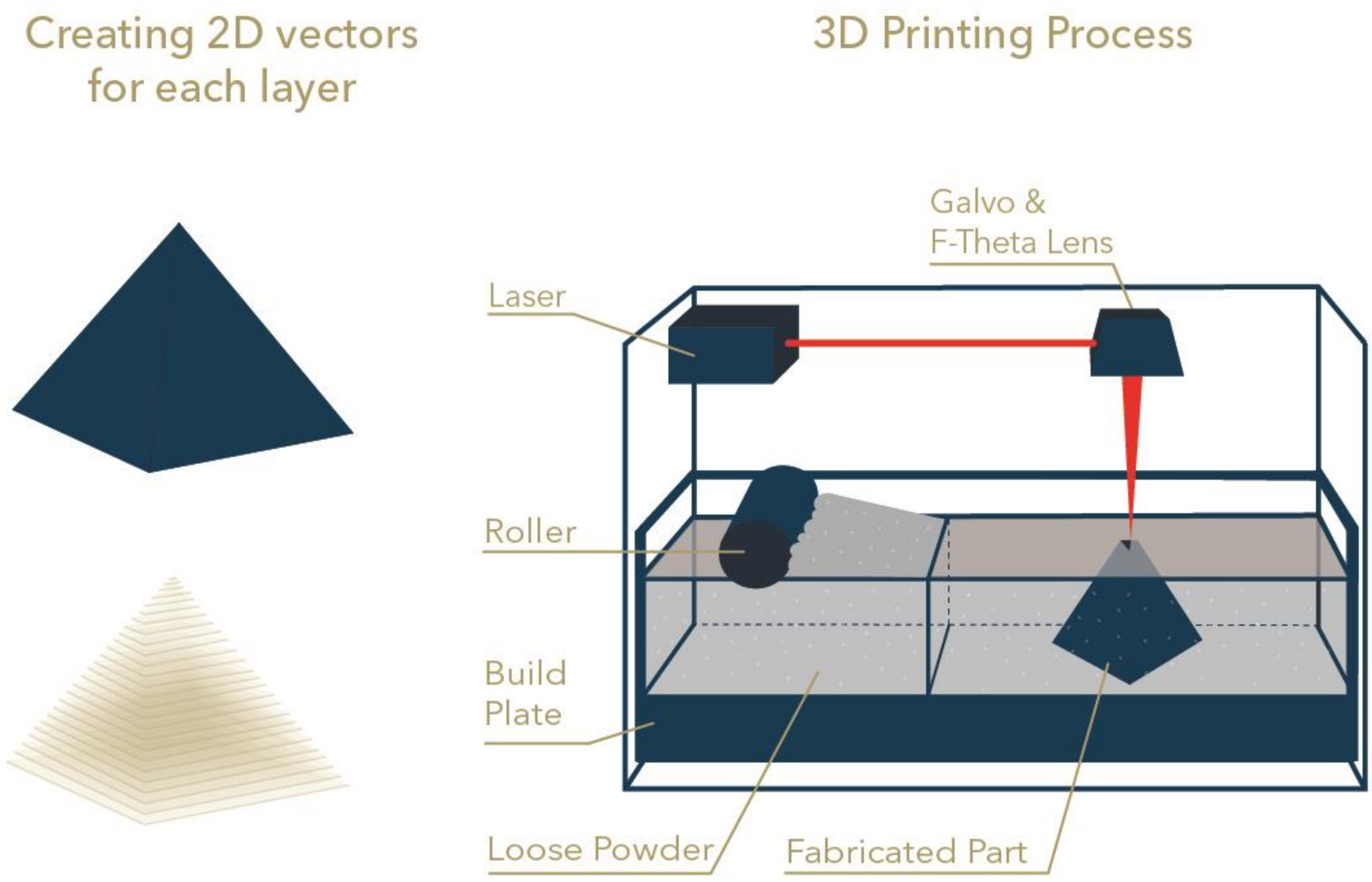
Device Manufacturing



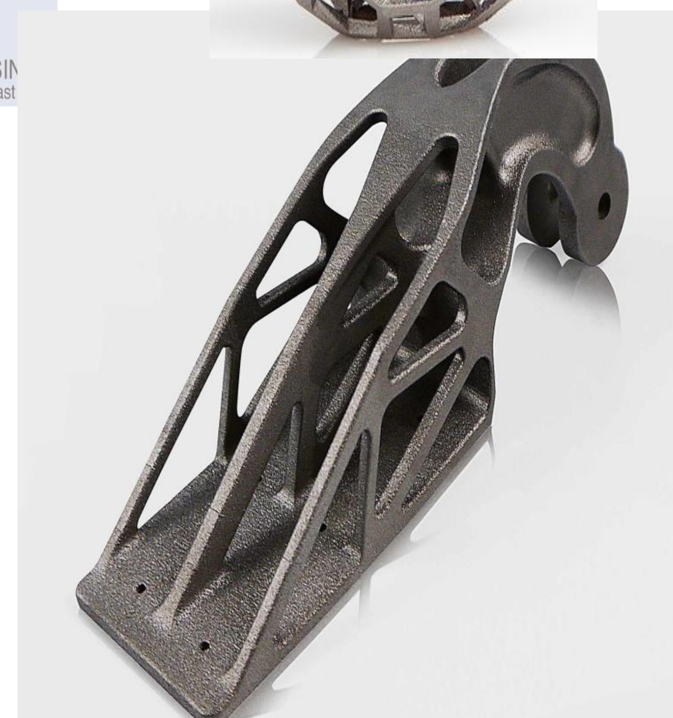
S. LeBlanc, "Thermoelectric Generators: Linking Material Properties and Systems Engineering for Waste Heat Recovery," *Sustainable Materials & Technologies* (2014)

Additive Manufacturing

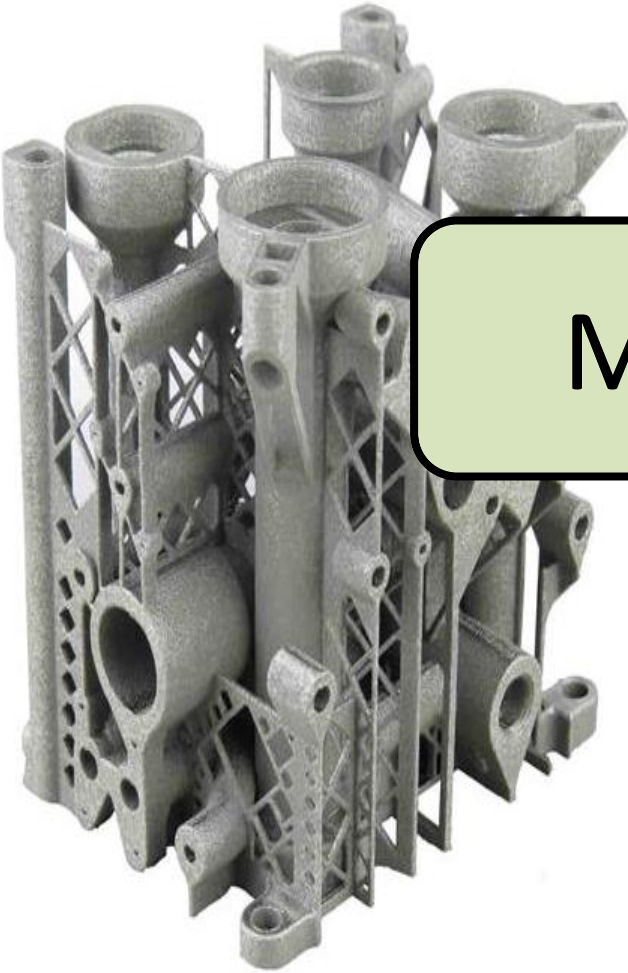
Laser Powder Bed Fusion:



Strong, Lightweight, Customized Parts

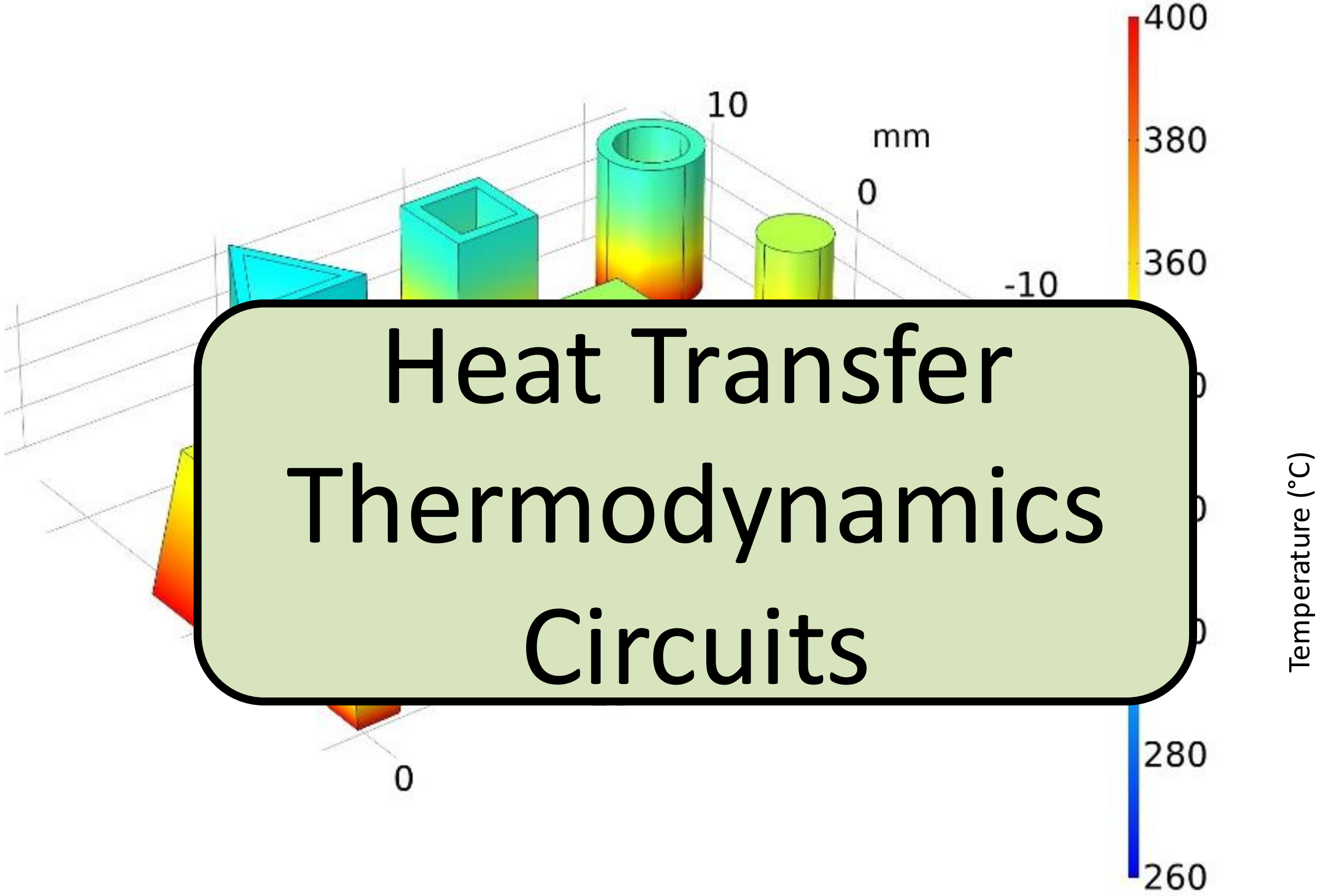


…with Small, Complex Features



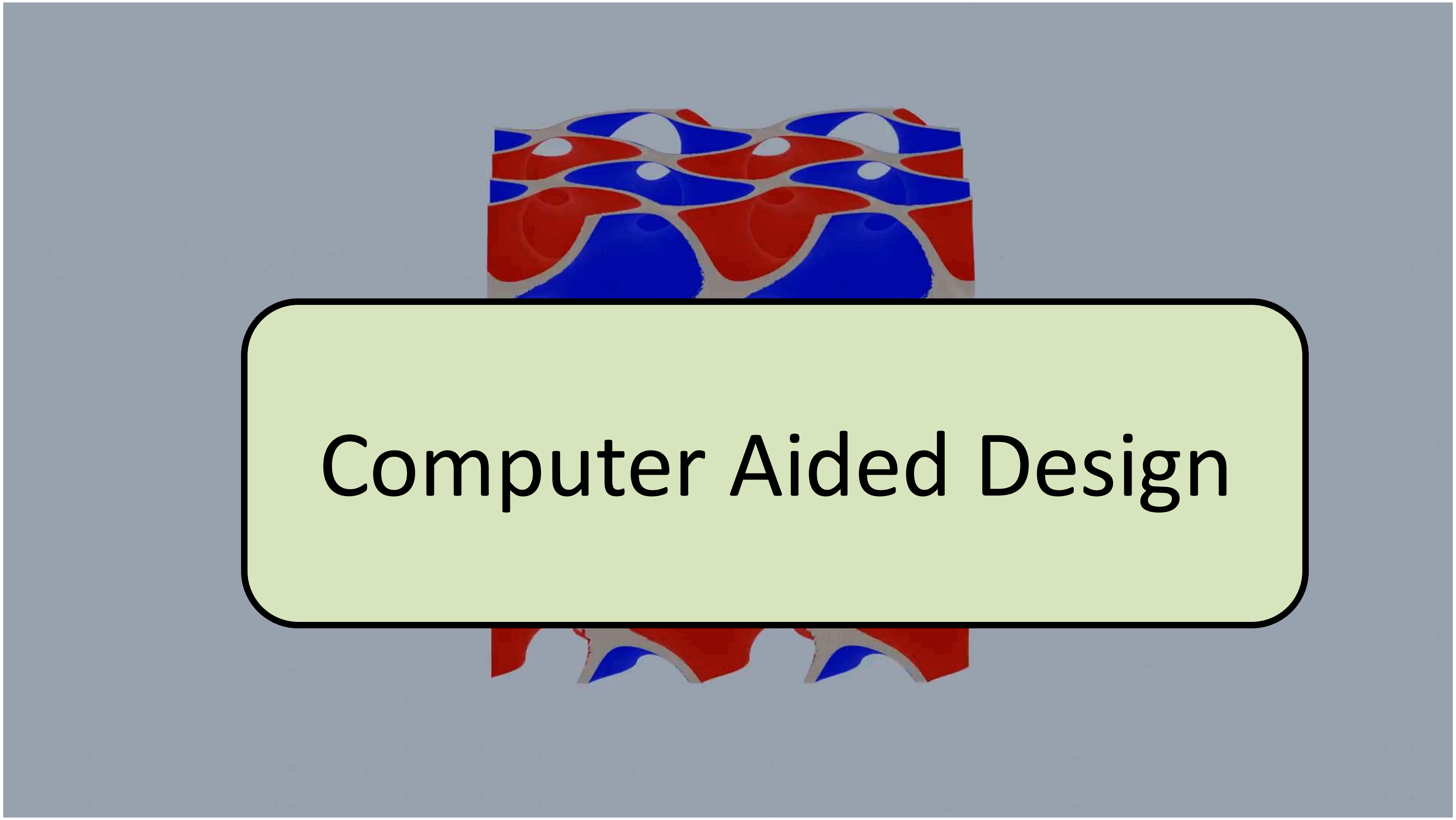
Manufacturing

Predicting Device Performance



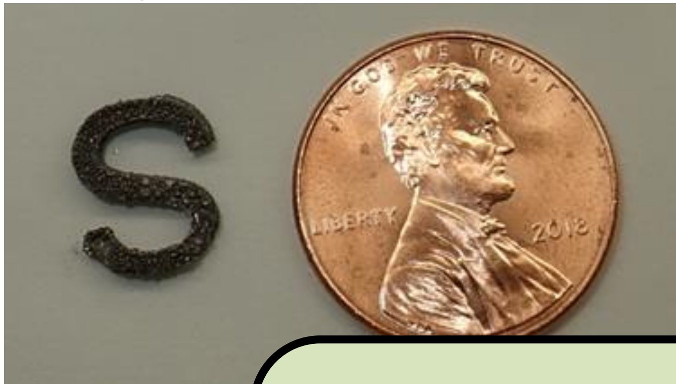
Heat Transfer
Thermodynamics
Circuits

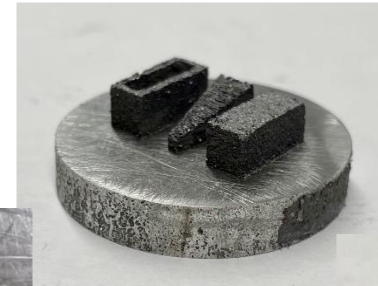
Device Design

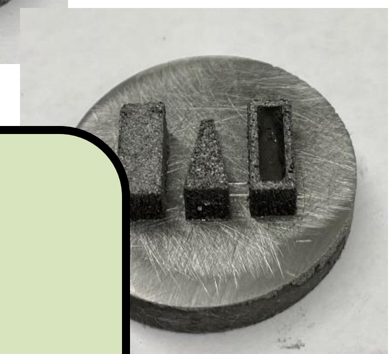


Prototypes

Laser powder bed fusion (selective laser melting):

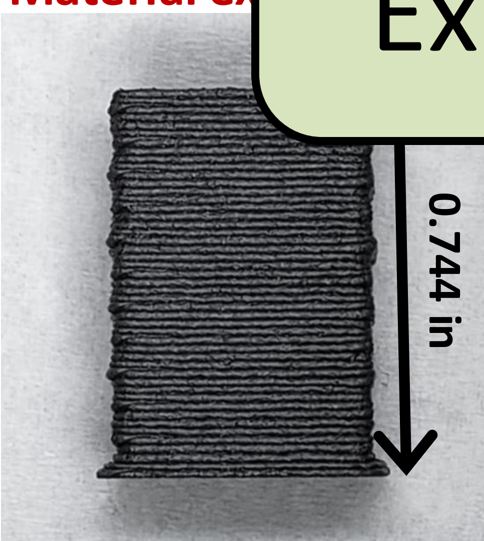


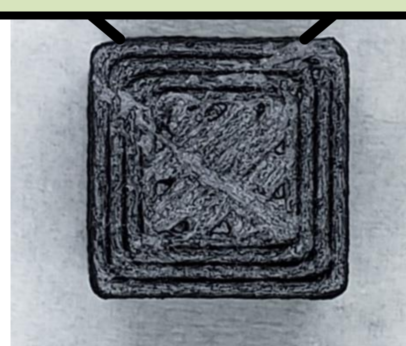




H. Zhang, S. LeB
& Performance,

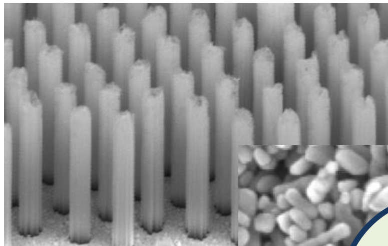
Material ex

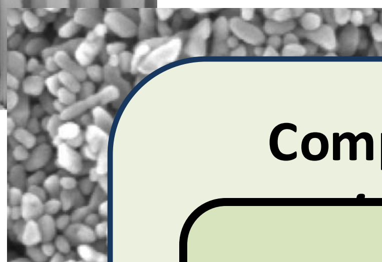


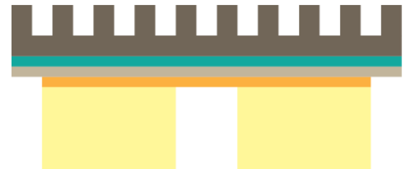


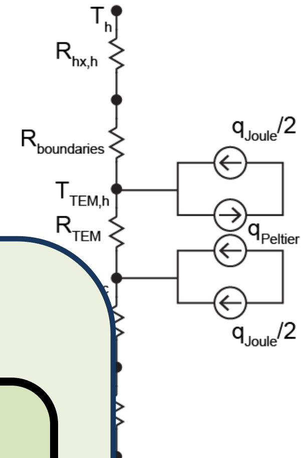
Mechanical Design
Experimental Methods

Cost-Performance Metric





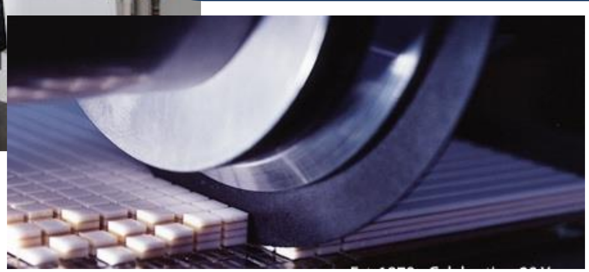




Comprehensive metric combining

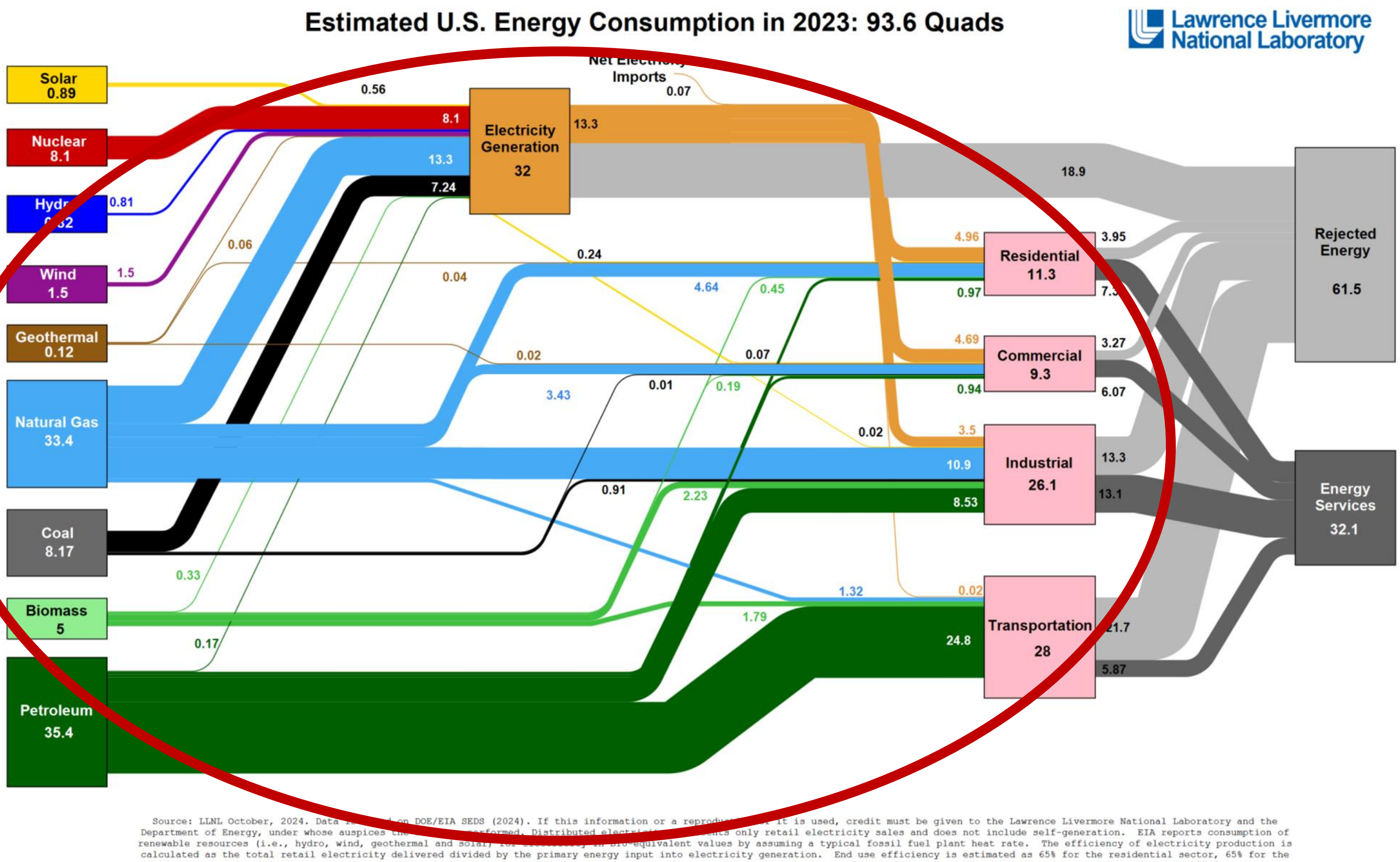
Engineering
Economics

$/Watt



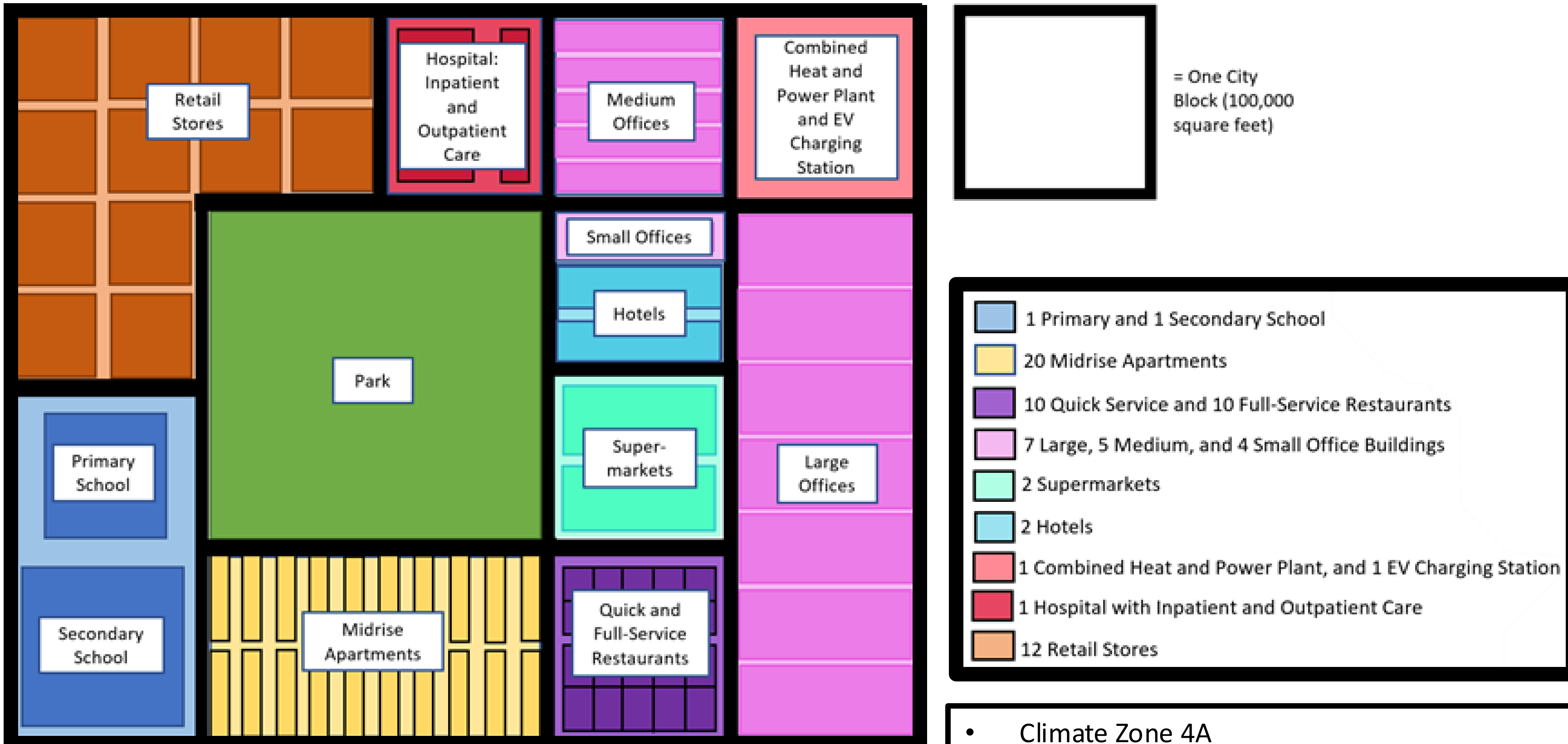
S. LeBlanc *et al.*, "Material and Manufacturing Cost Considerations for Thermoelectrics," *Renewable & Sustainable Energy Reviews* (2014)

How do we manage all that energy?



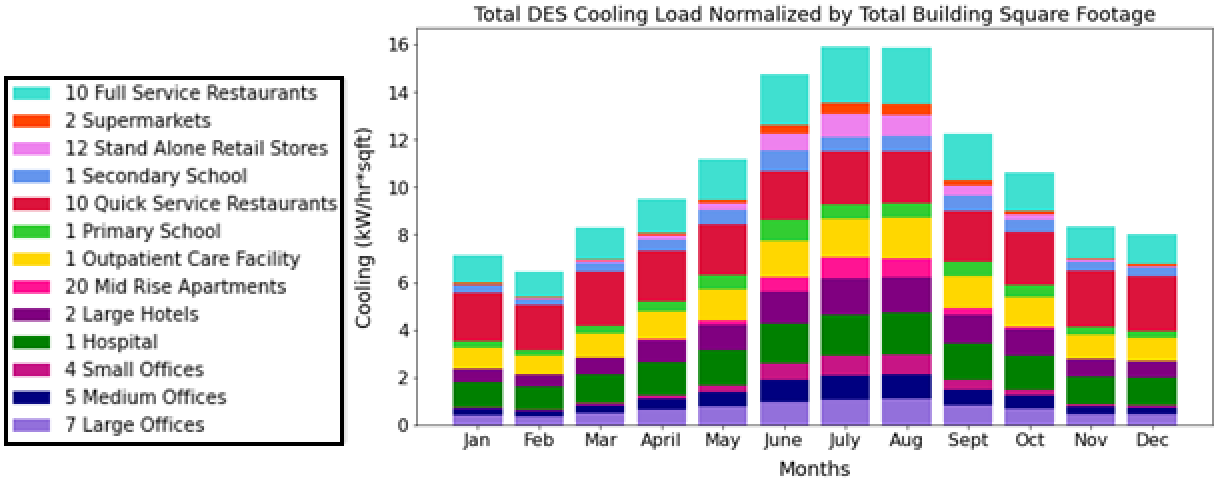
https://flowcharts.llnl.gov/commodities/energy

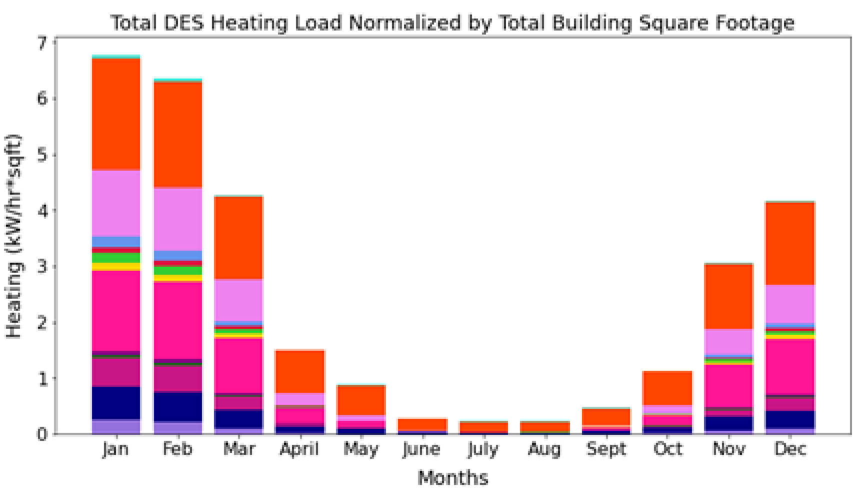
Example Urban District Energy System

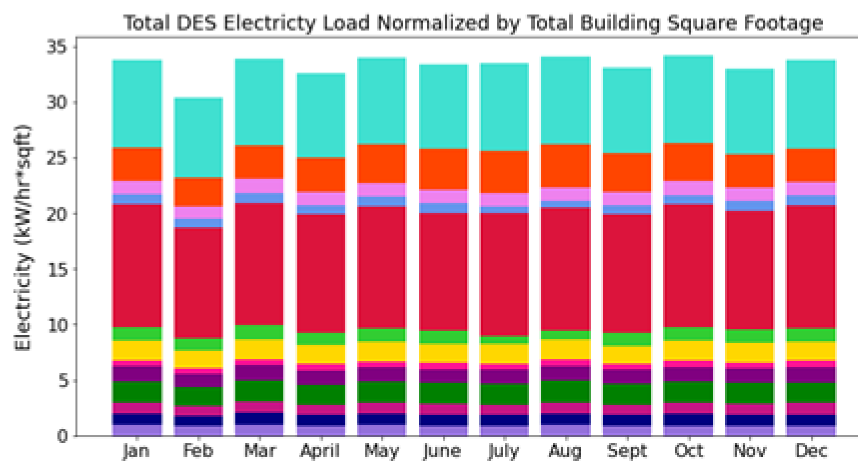


- Climate Zone 4A
- Combined Heat and Power Plant
- Hot and Chilled Water Thermal Storage
- Battery Storage
- Solar Photovoltaic Panels
- Electric Vehicle Charging Station

Energy Loads



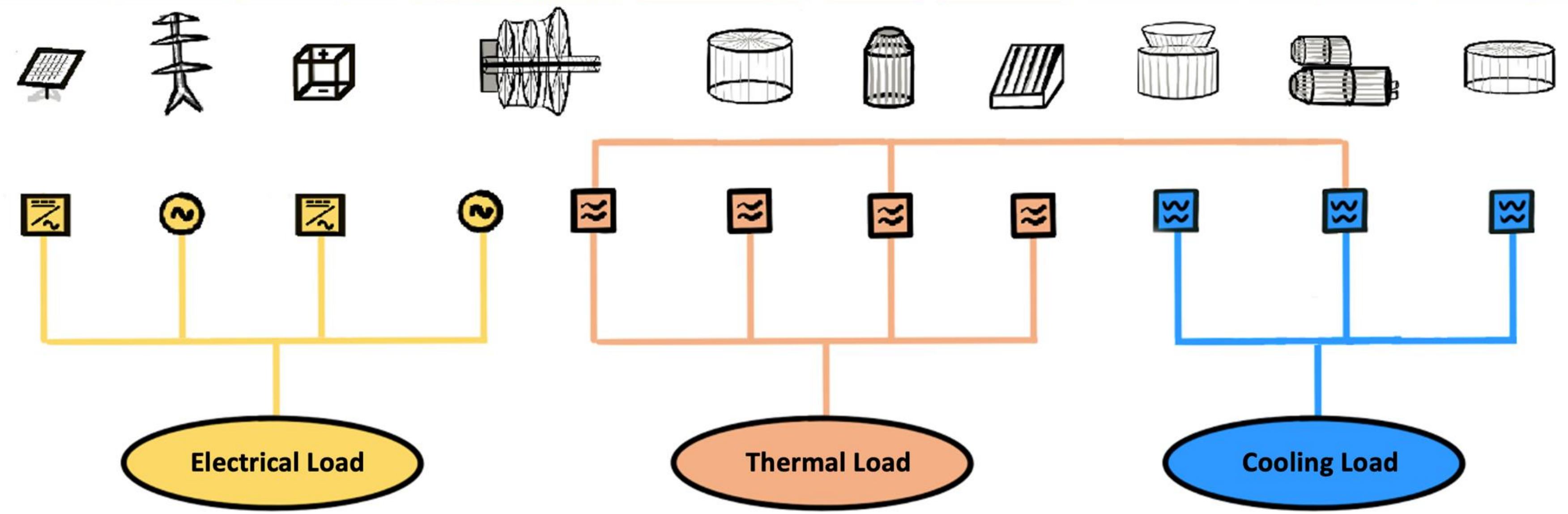




(normalized by building square footage)

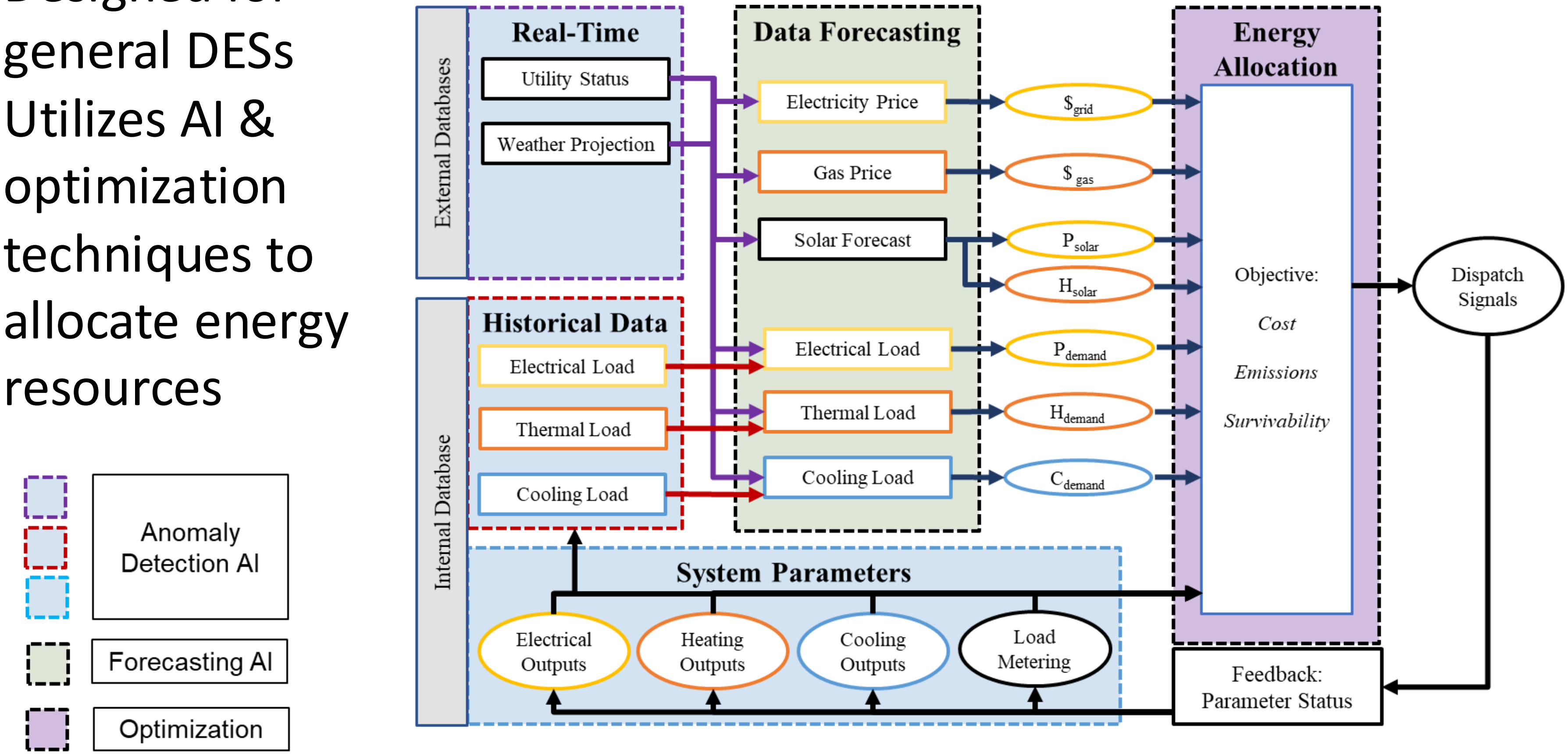
Energy Management System

	Solar	Grid	Battery	CHP	Thermal Storage	Boiler	Solar Heater	Towers & Chiller	Absorption Chiller	Ice Storage
Parameter	Power Output [kW]	Max In/Out [kW]	Capacity [kWh]	Heat Output [kW]	Capacity [kWh]	Heat Output [kW]	Heat Output [kW]	Output Max [gpm]	Output Max [gpm]	Capacity [gal]
	5491	5000	24071	1538	19569	23503	0	26024	2246	1754995
Sizing	Area [m^2]	Min In/Out [kW]	Critical Load Support [days]	Heat Load Percentile [%]	Critical Load Support [days]	% Peak Heat [%]	Area [m^2]	% Peak Cooling [%]	% CHP Peak Heat [%]	Critical Load Support [days]
	49139	0	0.50	75	0.50	125	0	125	100	0.50
Energy Resource	Solar	Grid	Battery	CHP	Thermal Storage	Boiler	Solar Heater	Towers & Chiller	Absorption Chiller	Ice Storage

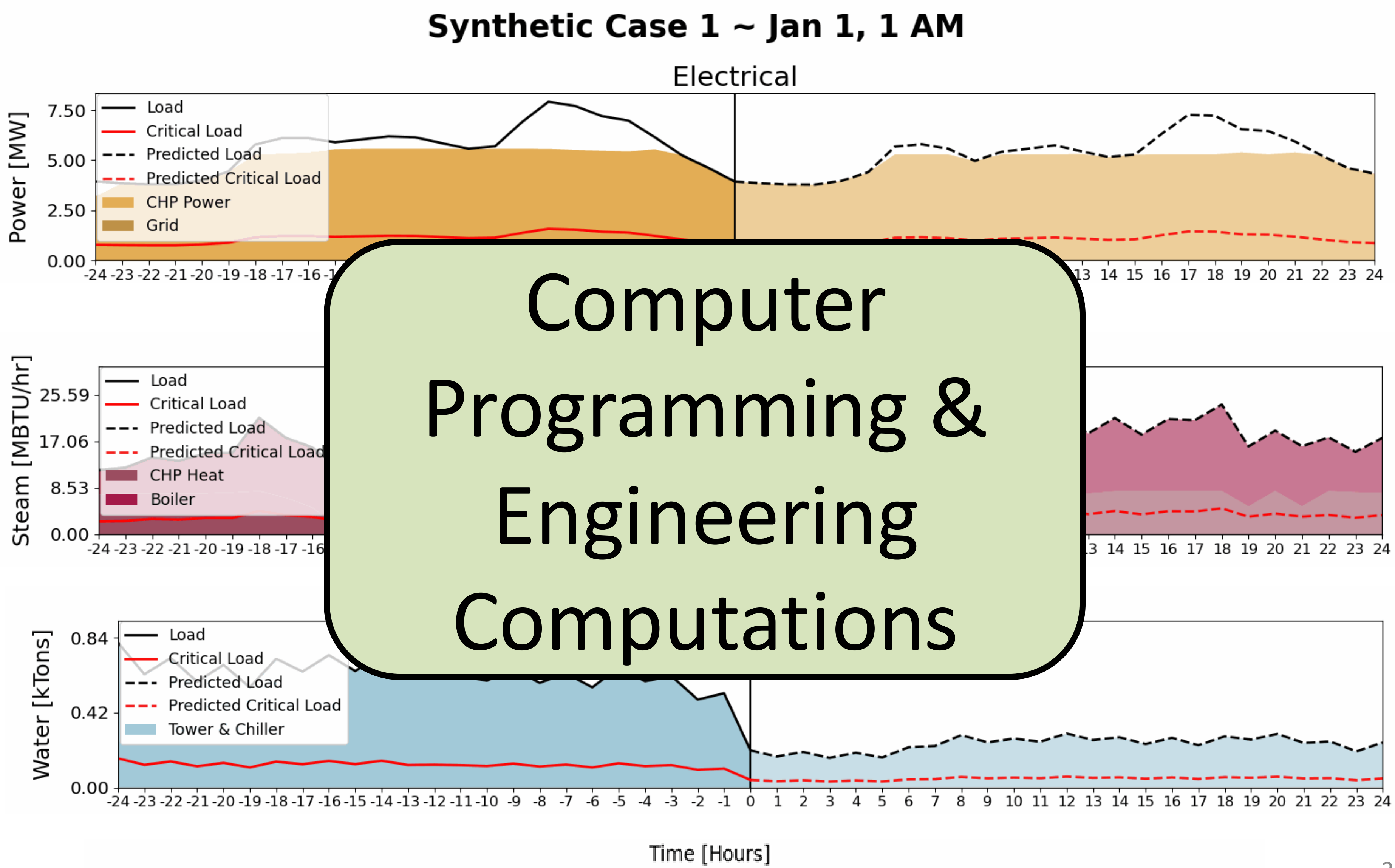


EMS Architecture

- Designed for general DESs
- Utilizes AI & optimization techniques to allocate energy resources

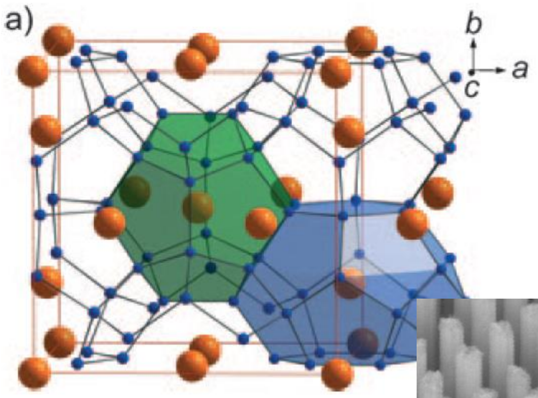


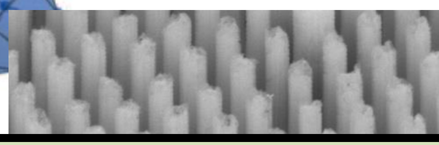
EMS Operation



Computer Programming & Engineering Computations

The Many Facets of Energy





Mechanical & Aerospace Engineering

Questions?

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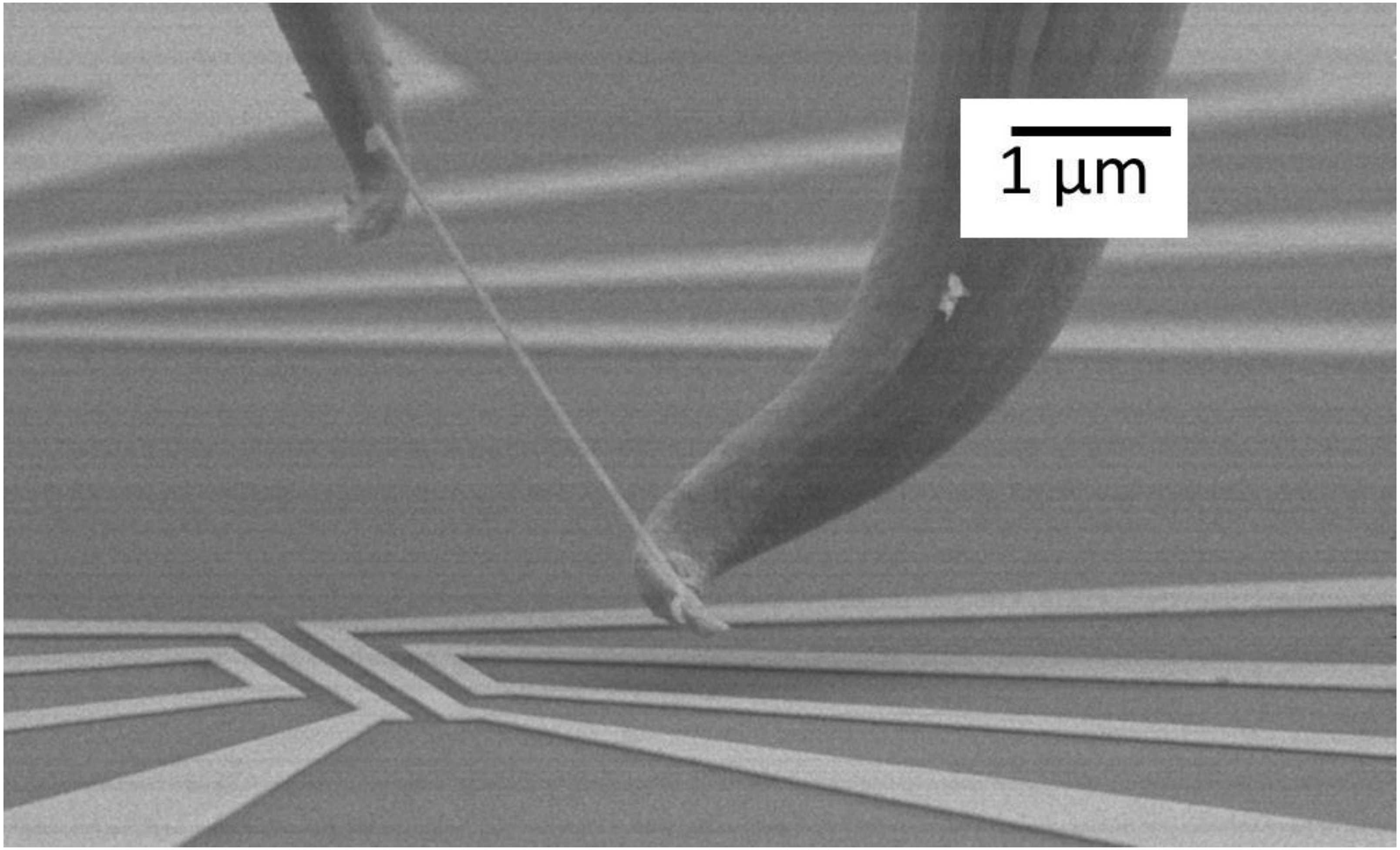

S. LeBlanc *et al., J. of Heat Transfer,* Photogallery, August 2012

Supported by DOE Center for Integrated Nanotechnologies